Assignment 8: Data Processing with Amazon Services (Online Retail)

UCI Online Retail transactions (541,909 lines, December 2010 to December 2011). The workflow extracts the data, applies CDC, cleans it with a Glue-style PySpark job, loads it into a warehouse, runs exploration SQL, and mines product association rules with Apriori.

1. What was run, and where

Stage	Built	Run here	Run on AWS
Extract to S3 (extract.py)	Yes	Yes, to a local S3 mirror	No: no AWS account
CDC (cdc.py, 03_cdc_merge.sql)	Yes	Yes, MERGE run in DuckDB	No
Glue ETL (transform.py, glue_etl_job.py)	Yes	Yes, transform on local Spark	No: awsglue not installed here
Redshift DDL, COPY, explore, mining SQL	Yes	Yes, in DuckDB (DISTKEY/SORTKEY removed)	No
Apriori and similarity (Python)	Yes	Yes	Not needed in AWS

The AWS parts are written to run as they are, but they were not run on AWS. Placeholders for account, role and bucket names must be filled in first. The DuckDB runs stand in for Redshift. Redshift itself was not used.

2. Extraction and stable keys

The workbook is read and written as CSV. Each line gets a *line_id* (invoice, product and occurrence number), assigned once at extraction. The natural key (InvoiceNo, StockCode) is not unique: 5,199 keys repeat. The line_id is what CDC matches on, so updates do not look like deletes followed by inserts.

3. Change data capture

A day-2 snapshot was simulated: about 1% of lines have their quantity changed, about 0.5% are deleted, and about 0.5% new lines are added. The diff on line_id gives the change set. The change set holds 2,709 inserts, 5,392 updates and 2,709 deletes. Applying it to the day-1 table with the MERGE in *sql/redshift/03_cdc_merge.sql* gives 541,909 rows. Every row matches the day-2 snapshot (all values match). The updates are fewer than the number selected, because some selected lines were then deleted.

4. Pre-processing and data quality

Step	Lines	Rule and reason
Raw input	541,909	All lines from the source
Cancellations	9,246	InvoiceNo starts with C; not a sale
Non-positive quantity	1,375	Returns and stock adjustments; baskets are built from purchases
Non-positive price	1,184	Price 0 or below (free items, write-offs, manual adjustments)
Non-product codes	2,347	POST, DOT, M, D, B, S, BANK CHARGES, gift vouchers and similar
Exact duplicates	5,098	Identical on every business column; smallest line_id kept
Clean lines	522,659	96.4% of raw lines kept

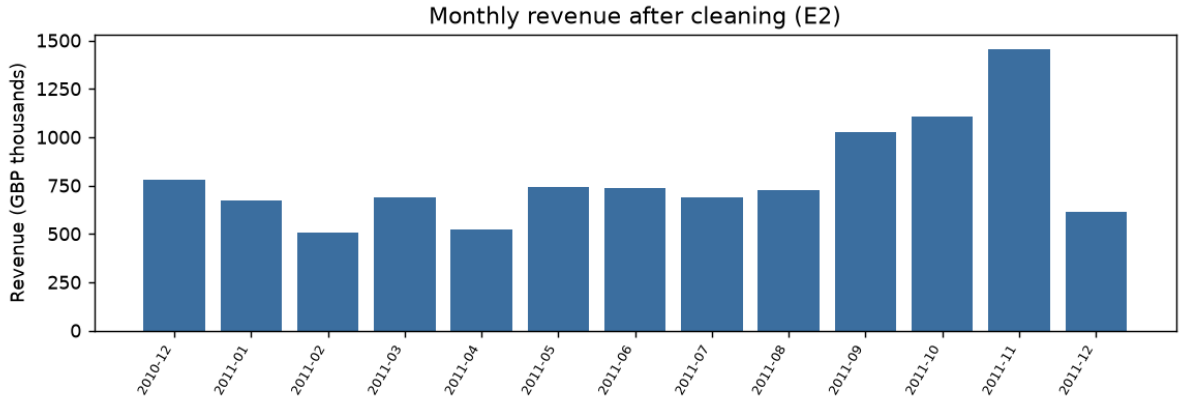
Missing values. 25.1% of clean lines have no CustomerID. These are guest checkouts. They are kept, flagged, and still count for basket mining, because a basket is an invoice and does not need a customer. Missing descriptions appear only on zero-price lines, so they are rejected above; the transform still fills and flags them if they occur.

Outliers. IQR fences (Quantity above 28.5; UnitPrice above 8.45) flag 26,805 quantity and 35,748 price lines. They are flagged, not removed: large wholesale quantities are real. The price fence is tight because prices are skewed, so the price flag is a review list, not an error list.

Sampling. The transform can take a stratified sample by country (sampleBy on Country). The reported results use the full clean data (sample fraction 1.0), because rules on a sample would have different support counts.

5. Exploration with SQL

The clean data covers 22,407 invoices, 4,333 known customers, 3,798 products and 38 countries, with revenue of GBP 10,264,656. The United Kingdom is the largest market.



Country	Revenue (GBP)	Invoices
United Kingdom	8,741,776	20,315
Netherlands	284,506	109
EIRE	271,221	322
Germany	205,844	482
France	184,642	416
Australia	135,580	60

Top product (E4)	Units	Revenue (GBP)
22423 REGENCY CAKESTAND 3 TIER	13,859	174,378
23843 PAPER CRAFT , LITTLE BIRDIE	80,995	168,470
85123A CREAM HANGING HEART T-LIGHT HOLDER	37,775	105,967
47566 PARTY BUNTING	18,083	98,605
85099B JUMBO BAG RED RETROSPOT	48,497	94,404

Products per invoice (E5)	Invoices
01-05	7,009
06-10	3,012
11-20	5,007
21-50	5,303
51+	2,076

6. Association rules (Apriori)

Basket = invoice (22,407 baskets), item = product code. Minimum support is 2% (at least 449 baskets), minimum confidence 50%, itemsets up to size 3. The Python Apriori and the SQL pair query were run on the same data. They found 54 frequent pairs each, and the pair counts are identical (yes). 291 frequent itemsets give 41 rules; 5 of them have a combined side.

Antecedent	Consequent	Baskets	Support	Confidence	Lift
22697 GREEN REGENCY TEACUP AND SAUCER + 22699 ROSES REGENCY TEACUP AND	22698 PINK REGENCY TEACUP AND SAUCER	528	0.024	0.70	20.5

Antecedent	Consequent	Baskets	Support	Confidence	Lift
22698 PINK REGENCY TEACUP AND SAUCER	22697 GREEN REGENCY TEACUP AND SAUCER + 22699 ROSES REGENCY TEACUP AND	528	0.024	0.69	20.5
22698 PINK REGENCY TEACUP AND SAUCER + 22699 ROSES REGENCY TEACUP AND	22697 GREEN REGENCY TEACUP AND SAUCER	528	0.024	0.89	19.8
22697 GREEN REGENCY TEACUP AND SAUCER	22698 PINK REGENCY TEACUP AND SAUCER + 22699 ROSES REGENCY TEACUP AND	528	0.024	0.52	19.8
22698 PINK REGENCY TEACUP AND SAUCER	22697 GREEN REGENCY TEACUP AND SAUCER	622	0.028	0.81	18.0
22697 GREEN REGENCY TEACUP AND SAUCER	22698 PINK REGENCY TEACUP AND SAUCER	622	0.028	0.61	18.0
22697 GREEN REGENCY TEACUP AND SAUCER + 22698 PINK REGENCY TEACUP AND	22699 ROSES REGENCY TEACUP AND SAUCER	528	0.024	0.85	17.9
23300 GARDENERS KNEELING PAD CUP OF TEA	23301 GARDENERS KNEELING PAD KEEP CALM	542	0.024	0.72	17.5
23301 GARDENERS KNEELING PAD KEEP CALM	23300 GARDENERS KNEELING PAD CUP OF TEA	542	0.024	0.59	17.5
22698 PINK REGENCY TEACUP AND SAUCER	22699 ROSES REGENCY TEACUP AND SAUCER	590	0.026	0.77	16.3

Reading the rules. Support is the share of baskets that contain both sides. Confidence is the share of baskets with the antecedent that also contain the consequent. Lift is confidence divided by the consequent's overall support; above 1 means the two products are bought together more than chance predicts. The strongest rules are the three Regency teacup and saucer colours, so most of the strong rules describe one bundle in three colours. Sixteen rules have lift above 15. They involve the three teacup colours, the Spaceboy and Dolly Girl lunch boxes, the Gardeners Kneeling Pad pair and the two Charlotte bags. These look like products that customers buy together, which suggests cross-selling. A high lift on a bundle is expected and does not by itself show a new recommendation.

7. Similarity and dissimilarity

Each product is a set of baskets (invoices). For two products A and B: **Jaccard** = |A and B| / |A or B| (overlap relative to everything either product appears in). **Cosine** = |A and B| / sqrt(|A| |B|), which rewards overlap without penalising very popular products as much. **Dissimilarity** = 1 minus Jaccard, a distance between products: 0 means always bought together, 1 means never together. Values are for the 20 most frequent products (190 pairs).

Product A	Product B	Co-baskets	Jaccard	Cosine	Dissimilarity
JUMBO BAG PINK POLKADOT	JUMBO BAG RED RETROSPOT	818	0.328	0.512	0.672
LUNCH BAG BLACK SKULL.	LUNCH BAG SUKI DESIGN	579	0.293	0.454	0.707
LUNCH BAG RED RETROSPOT	LUNCH BAG SUKI DESIGN	640	0.290	0.452	0.710
LUNCH BAG RED RETROSPOT	LUNCH BAG BLACK SKULL.	637	0.289	0.450	0.711
JUMBO STORAGE BAG SUKI	JUMBO SHOPPER VINTAGE RED PAISLEY	518	0.283	0.441	0.717
JUMBO STORAGE BAG SUKI	JUMBO BAG RED RETROSPOT	718	0.281	0.457	0.719

The most similar pairs are carrier bags: the lunch bag and jumbo bag designs are bought in the same invoices. The least similar pair in the top 20 has a Jaccard of 0.031 (dissimilarity 0.969), so the top products do not all share baskets. Jaccard is the main measure; cosine is never below Jaccard, and the two rank the 190 pairs almost identically (Spearman 0.999), so either can drive a clustering step. Dissimilarity (1 minus Jaccard) is the distance such a step would use.

8. Limitations and next steps

• No AWS services were used. The Glue job has not been run in Glue, and the Redshift SQL has not been run on a Redshift cluster. Both run here on local stand-ins with the same logic.

• The day-2 change set is simulated. A real source would supply its own change log.

• The minimum support (2%) and confidence (50%) were chosen, not tuned. A sweep over support is the next step.

• Basket mining uses all clean invoices. Guest checkouts (about a quarter of lines) are included.

• Screenshots of the AWS consoles are not included, because nothing was deployed.

9. References

• Chen, D. Online Retail. UCI Machine Learning Repository, 2015. https://archive.ics.uci.edu/dataset/352/online+retail

• Agrawal, R. and Srikant, R. Fast algorithms for mining association rules. VLDB 1994.

• Tan, P.-N., Steinbach, M. and Kumar, V. Introduction to Data Mining. Pearson, 2nd ed. (Jaccard and cosine).

• Amazon Redshift Database Developer Guide: MERGE, COPY, distribution and sort keys.

• AWS Glue Developer Guide: Glue 4.0 jobs, job bookmarks and --extra-py-files.